# Hypothesis Testing

**ANSWER KEY**

## Section 1: Null & Alternative Hypotheses, Type I & Type II Errors

**1** Define a Type I error in hypothesis testing.

**a.** **Rejecting $H_0$ when $H_0$ is actually true ✓**

**b.** Failing to reject $H_0$ when $H_0$ is false

**c.** Setting the significance level too high

**d.** Accepting $H_1$ when it is false

A Type I error occurs when a true null hypothesis is incorrectly rejected. The probability of a Type I error equals the significance level $\alpha$.

**2** Define a Type II error in hypothesis testing.

**a.** **Failing to reject $H_0$ when $H_0$ is false ✓**

**b.** Rejecting $H_0$ when $H_0$ is true

**c.** Computing an incorrect test statistic

**d.** Rejecting $H_1$ when $H_0$ is true

A Type II error (probability $\beta$) occurs when the test fails to reject a false null hypothesis.

**3** A machine produces metal pins with mean length 5.0 cm. A technician suspects the machine is out of adjustment and producing pins of different mean length. What are the appropriate hypotheses?

**a.** $H_0 : \mu = 5.0$, $H_1 : \mu \neq 5.0$ ✓

**b.** $H_0 : \mu = 5.0, H_1 : \mu > 5.0$

**c.** $H_0 : \mu \neq 5.0, H_1 : \mu = 5.0$

**d.** $H_0 : \mu = 5.0, H_1 : \mu < 5.0$

A non-directional suspicion ('different mean length') requires a two-tailed alternative hypothesis $H_1 : \mu \neq 5.0$.

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## Section 2: Testing Population Mean with Normal Distribution

**4** A test of $H_0 : \mu = 80$ against $H_1 : \mu > 80$ at the $5\%$ level is conducted. A sample of $n = 36$ from $N(\mu, 144)$ gives $\bar{x} = 83.5$. What is the value of the test statistic $z$?

**a.** $z = 1.75$ ✓

**b.** $z = 1.645$

**c.** $z = 2.00$

**d.** $z = 0.875$

$\sigma = 12$, $\text{SE} = 12/\sqrt{36} = 2.0$. Test statistic $z = \frac{83.5 - 80}{2.0} = \frac{3.5}{2.0} = 1.75$.

**5** For the test in the previous question ($z = 1.75$), what is the test conclusion at the $5\%$ significance level (critical value $1.645$)?

**a.** **Reject $H_0$; there is sufficient evidence that the mean is greater than 80** ✓

**b.** Fail to reject $H_0$; insufficient evidence

**c.** Accept $H_0$; the difference is due to chance

**d.** Test is inconclusive

Since $z = 1.75 > 1.645$, the test statistic falls in the critical region. Reject $H_0$ at the 5% level.

**6** What is the $p$-value for the test statistic $z = 1.75$ in a one-tailed test?

**a.** $0.0401$ ✓

**b.** $0.0802$

**c.** $0.9599$

**d.** $0.0250$

$p\text{-value} = 1 - \Phi(1.75) = 1 - 0.9599 = 0.0401$.

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## Section 3: Testing Parameters with Poisson & Binomial Models

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**7** A company claims its weekly defect rate is $\lambda = 2$. In a given week, 6 defects are observed. Test $H_0 : \lambda = 2$ against $H_1 : \lambda > 2$ by calculating $P(X \geq 6)$ under $H_0$.

**a.** $0.0166$ ✓

**b.** $0.0527$

**c.** $0.0345$

**d.** $0.0045$

For Po(2):
$P(X \leq 5) = e^{-2}(1 + 2 + 2 + 4/3 + 2/3 + 4/15) = e^{-2}(109/15) = 0.135335 \times 7.2667 \approx 0.9834$.
$P(X \geq 6) = 1 - 0.9834 = 0.0166$.

**8** At the $5\%$ significance level, what is the conclusion for the defect test in the previous question ( $p = 0.0166$)?

**a.** **Reject $H_0$; the defect rate has significantly increased** ✓

**b.** Fail to reject $H_0$

**c.** Accept $H_0$

**d.** Change the significance level to $1\%$

Since $p = 0.0166 < 0.05$, reject $H_0$ at the 5% level.

**9** A seed packet claims a $90\%$ germination rate. A gardener tests 20 seeds and finds 15 germinate. To test $H_0 : p = 0.9$ against $H_1 : p < 0.9$, which probability must be computed?

**a.** $P(X \leq 15)$ **where** $X \sim B(20, 0.9)$ ✓

**b.** $P(X = 15)$ where $X \sim B(20, 0.9)$

**c.** $P(X \geq 15)$ where $X \sim B(20, 0.9)$

**d.** $P(X \leq 15)$ where $X \sim N(18, 1.8)$

For a lower-tail test ($H_1 : p < 0.9$), the $p$-value is the probability of observing an outcome as extreme as or more extreme than the observed value: $P(X \leq 15)$.

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## Section 4: Critical Regions & Power of a Test

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**10** For a test of $H_0 : \mu = 50$ vs $H_1 : \mu < 50$ using $n = 25, \sigma = 10$, find the critical value for $\bar{x}$ at the 5% level ( $z_c = -1.645$).

**a.** 46.71 ✓

**b.** 45.00

**c.** 47.50

**d.** 46.00

Critical value $= \mu_0 - z_c \frac{\sigma}{\sqrt{n}} = 50 - 1.645\left(\frac{10}{5}\right) = 50 - 1.645(2) = 50 - 3.29 = 46.71.$

**11** In the previous test (critical region $\bar{X} < 46.71$), what is the probability of a Type II error if the true mean is $\mu = 45$?

**a.** 0.1964 ✓

**b.** 0.8036

**c.** 0.0500

**d.** 0.3085

Type II error $\beta = P(\bar{X} \geq 46.71 \mid \mu = 45)$. Under $\mu = 45$: $z = \frac{46.71 - 45}{2.0} = 0.855$.
$\beta = 1 - \Phi(0.855) = 1 - 0.8037 = 0.1963 \approx 0.1964.$

**12** What is the statistical power of the test in the previous question (where $\beta = 0.1964$)?

**a.** 0.8036 ✓

**b.** 0.1964

**c.** 0.9500

**d.** 0.0500

Power of a test is defined as $1 - \beta = 1 - 0.1964 = 0.8036$ (or 80.36%).

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